

De binario a decimal

Recuerda: Cada dígito binario (bit) representa una potencia de 2, empezando desde 2^0 (1) para el bit más a la derecha. Multiplica cada bit por su valor correspondiente y suma los resultados.

1. **1011:** 8 4 2 1

$$8 \times 1 = 8 \mid 4 \times 0 = 0 \mid 2 \times 1 = 2 \mid 1 \times 1 = 1$$

$$8 + 0 + 2 + 1 = 11$$

2. **11001:** 16 8 4 2 1

$$16 \times 1 = 16 \mid 8 \times 1 = 8 \mid 4 \times 0 = 0 \mid 2 \times 0 = 0 \mid 1 \times 1 = 1$$

$$16 + 8 + 0 + 0 + 1 = 25$$

3. **101010:** 32 16 8 4 2 1

$$32 \times 1 = 32 \mid 16 \times 0 = 0 \mid 8 \times 1 = 8 \mid 4 \times 0 = 0 \mid 2 \times 1 = 2 \mid 1 \times 0 = 0$$

$$32 + 0 + 8 + 0 + 2 + 0 = 42$$

4. **1110101:** 64 32 16 8 4 2 1

$$64 \times 1 = 64 \mid 32 \times 1 = 32 \mid 16 \times 1 = 16 \mid 8 \times 0 = 0 \mid 4 \times 1 = 4 \mid 2 \times 0 = 0 \mid 1 \times 1 = 1$$

$$64 + 32 + 16 + 0 + 4 + 0 + 1 = 117$$

5. **11100110:** 128 64 32 16 8 4 2 1

$$128 \times 1 = 128 \mid 64 \times 1 = 64 \mid 32 \times 1 = 32 \mid 16 \times 0 = 0 \mid 8 \times 0 = 0 \mid 4 \times 1 = 4 \mid 2 \times 1 = 2 \mid 1 \times 0 = 0$$

$$128 + 64 + 32 + 0 + 0 + 4 + 2 + 0 = 230$$

6. **10011101:** 128 64 32 16 8 4 2 1

$$128 \times 1 = 128 \mid 64 \times 0 = 0 \mid 32 \times 0 = 0 \mid 16 \times 1 = 16 \mid 8 \times 1 = 8 \mid 4 \times 1 = 4 \mid 2 \times 0 = 0 \mid 1 \times 1 = 1$$

$$128 + 0 + 0 + 16 + 8 + 4 + 0 + 1 = 157$$

De decimal a binario

Recuerda: Divide el número decimal entre 2 repetidamente hasta obtener un cociente de 0. Los restos de cada división, leídos de abajo hacia arriba, formarán el número binario.

1. **13:** $13 / 2 = 6$ remainder **1**
 $6 / 2 = 3$ remainder **0**
 $3 / 2 = 1$ remainder **1**
 $1 / 2 = 0$ remainder **1**

13 = 1101

2. **22:** $22 / 2 = 11$ remainder **0**
 $11 / 2 = 5$ remainder **1**
 $5 / 2 = 2$ remainder **1**
 $2 / 2 = 1$ remainder **0**
 $1 / 2 = 0$ remainder **1**

22 = 10110

3. **57:** $57 / 2 = 28$ remainder **1**
 $28 / 2 = 14$ remainder **0**
 $14 / 2 = 7$ remainder **0**
 $7 / 2 = 3$ remainder **1**
 $3 / 2 = 1$ remainder **1**
 $1 / 2 = 0$ remainder **1**

57 = 111001

4. **100:** $100 / 2 = 50$ remainder **0**
 $50 / 2 = 25$ remainder **0**
 $25 / 2 = 12$ remainder **1**
 $12 / 2 = 6$ remainder **0**
 $6 / 2 = 3$ remainder **0**
 $3 / 2 = 1$ remainder **1**
 $1 / 2 = 0$ remainder **1**

100 = 1100100

5. **124:** $124 / 2 = 62$ remainder **0**
 $62 / 2 = 31$ remainder **0**
 $31 / 2 = 15$ remainder **1**
 $15 / 2 = 7$ remainder **1**

$$7 / 2 = 3 \text{ remainder } 1$$

$$3 / 2 = 1 \text{ remainder } 1$$

$$1 / 2 = 0 \text{ remainder } 1$$

$$124 = 1111100$$